

Food Delivery Platform Analytics

End-to-End SQL Analytics on 2.5M+ Records | Indian
Food Delivery Market

Palagiri Shashank Reddy

B.Tech CSE Graduate — VIT Vellore, 2026

SQLite

Python

Tableau

Pandas

Seaborn

Executive Summary

Project Overview

This project built an end-to-end SQL analytics platform analyzing 2.5M+ records across a 9-table relational SQLite database. It simulates large-scale food delivery operations across 9 Indian cities, transforming raw operational data into actionable business intelligence using advanced SQL techniques and data visualization.

INR 337M

TOTAL GMV (2022-2024)

425,313

DELIVERED ORDERS

2.5M+

DATA RECORDS

9

RELATIONAL TABLES

85.55%

ON-TIME DELIVERY RATE

9

INDIAN CITIES COVERED

Business Objectives

- **Revenue & Growth:** Identify top revenue-generating cities, track month-over-month revenue trends, and pinpoint restaurants contributing most to GMV.
- **Customer Analytics:** Detect customers at high risk of churning, segment customers by lifetime value, and evaluate the effectiveness of promotional campaigns on retention.
- **Operations:** Monitor delivery performance against promised timelines, analyze system stress during peak hours, and evaluate the performance of delivery partners.

Database Architecture

The analytics platform operates on a robust 9-table relational schema designed to capture every aspect of the food delivery lifecycle.

**Note: Restaurant data leverages real Swiggy data from Kaggle, while transactional and customer records are synthetically generated to model real-world scale and distributions.*

Table Name	Row Count	Description
restaurants	8,680	Core entity containing real Swiggy restaurant profiles.
customers	50,000	User profiles and demographic information.
orders	500,000	Header table for all transactional events.
order_items	1,250,000	Line-item details mapping menus to orders.
delivery_partners	2,000	Fleet personnel information and ratings.
delivery_tracking	425,644	Logistical timestamps for route optimization.
reviews	255,386	Customer feedback and rating aggregations.
menu	65,179	Item catalog linked to respective restaurants.
promotions	50	Campaign parameters and discount configurations.

Data Sources & Processing

The foundation of this analysis relies on a hybrid dataset, combining real-world entity data (Swiggy restaurant database from Kaggle) with procedurally generated transactional events (2M+ rows) designed to simulate the statistical properties of enterprise food delivery operations. The data was structured into an optimized SQLite database with 11 strategic indexes to ensure performant complex queries.

Technology Stack

Technology	Purpose in Project
SQLite	Primary database engine; executing complex aggregations.
Python	Data ingestion, orchestration, and pipeline automation.
Pandas & NumPy	In-memory data manipulation and statistical structuring.
Tableau	Building interactive business intelligence dashboards.
Seaborn/Matplotlib	Exploratory data analysis and static visualizations.
Jupyter Notebook	Interactive environment for analysis modules.

Analysis Modules

The analysis was structured into key business modules: Cohort Retention Analysis, RFM (Recency, Frequency, Monetary) Segmentation, Delivery Operations Tracking, Revenue Concentration Profiling, and Campaign Effectiveness Evaluation.

Key Business Findings

1. Total Platform Volume

INR 337M GMV

The platform processed 425,313 successfully delivered orders over a 2-year period (2022-2024).

Recommendation: Continue scaling infrastructure to support volume growth.

2. City Performance Disparity

~44% GMV from Top 3 Cities

Kolkata (INR 53.7M) is the highest performer, followed by Mumbai (INR 48.2M) and Chennai (INR 46.0M).

Recommendation: Prioritize marketing and operational spend in top-performing metros.

3. Operational Excellence

85.55% On-Time Delivery

Consistently maintained across all 9 cities, well above the 75% industry standard benchmark.

Recommendation: Use this metric as a core value proposition in customer marketing.

Key Business Findings (Cont.)

5. Promo Effectiveness

99.9% Repeat Purchasing

Customers utilizing promotions show a higher repeat purchasing rate (99.9%) compared to non-promo customers (98.7%).

Recommendation: Leverage targeted promotions specifically for retention.

6. Peak Hour Bottlenecks

9% Slower Deliveries

During the 7-10 PM peak period, delivery times degrade and take ~9% longer than non-peak deliveries.

Recommendation: Pre-position fleet partners based on historical peak demand maps.

7. Customer Churn Segmentation

~7,000 At-Risk Customers

While 42,000+ customers fall into the Low-Risk segment, roughly 7,000 require immediate retention attention.

Recommendation: Deploy targeted win-back campaigns for the at-risk segment.

Strategic Business Recommendations

1. Volume-Quality Recovery Programme

Problem Solved: Protecting GMV from high-volume but poorly rated vendors.

Data Evidence: High revenue concentration in the top restaurant quartile.

Proposed Action: Mandatory food quality audits and packaging reviews for high-volume restaurants rated below 3.5.

Expected Impact: Increased order retention and higher customer satisfaction in high-throughput nodes.

2. Real-Time Fleet Surge Alerts

Problem Solved: Service degradation during evening dinner rushes.

Data Evidence: 7-10 PM deliveries take ~9% longer than non-peak hours.

Proposed Action: Deploy automated, proactive partner alerts 45 minutes before demand peaks.

Expected Impact: Improved on-time metrics during peak hours and better load balancing across the fleet.

3. Promo Re-engagement Funnel

Problem Solved: Mitigating revenue loss from the at-risk customer segment.

Data Evidence: ~7,000 customers require retention attention; promos lift repeat rates to 99.9%.

Proposed Action: Target the 30-60 day dormancy window with specific discount campaigns.

SQL Concepts Demonstrated

The analytics platform utilized advanced SQL capabilities to extract multi-dimensional insights from the relational structure.

SQL Concept	Business Application
Common Table Expressions (CTEs)	Simplifying complex multi-step revenue aggregations and funnel analysis.
Window Functions (LAG, NTILE, RANK)	Calculating month-over-month growth, quartile segmentations, and ranking top performers.
Cohort Retention Analysis	Tracking user engagement decay over time to inform marketing spend.
RFM Segmentation	Classifying the customer base by Recency, Frequency, and Monetary value for targeted campaigns.
Running Totals	Monitoring cumulative GMV and order volumes across fiscal periods.
Correlated Subqueries	Benchmarking individual restaurant performance against city-wide averages.
Complex JOIN Chains	Connecting 3-4 tables (e.g., Orders, Items, Menu, Restaurants) to trace line-item revenue.
HAVING Clause Filtering	Isolating significant data groups, such as users with high order frequencies.
Index Optimization	Deploying 11 strategic indexes to drastically reduce query execution time on 2.5M+ records.
Date Functions (julianday, strftime)	Extracting time-of-day for peak analysis and calculating exact delivery duration arithmetic.

Project Links & Author

Project Repository & Dashboards

GitHub Repository: github.com/Shanksreddy005/food-delivery-analytics-sql

Kaggle Notebook: kaggle.com/code/shashankreddy123987/food-delivery-analytics-advanced-sql-case-study

Tableau Dashboard: public.tableau.com/app/profile/shashank.reddy4125

LinkedIn Profile: linkedin.com/in/shashank-reddy-147227260

About the Author

Palagiri Shashank Reddy

B.Tech CSE Graduate — VIT Vellore, Class of 2026

A data-driven professional specializing in SQL analytics, data engineering, and business intelligence. Passionate about transforming raw data sets into strategic insights that drive operational efficiency and revenue growth.

"Data is only as valuable as the decisions it enables."